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Patch-Based Image Focus Detection

A Comparison of Classical Machine Learning and Lightweight Neural
Networks

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Exposé for a Bachelor's Thesis

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1 Problem Statement

Image focus detection aims to distinguish focused (sharp) regions from unfocused (blurred) regions within an image and is widely used in image analysis and quality assessment. A common strategy is to segment images into regions and classify these regions according to their focus characteristics.

Traditional machine learning approaches for this task rely on handcrafted feature extraction followed by supervised classification. Although such methods are computationally efficient, their robustness is often limited by the expressiveness of manually designed features, and they may suffer from overfitting when image conditions vary. Convolutional neural networks (CNNs), in contrast, are able to automatically learn discriminative features and generally exhibit stronger robustness. However, this capability typically comes at the cost of increased data requirements, making CNN-based methods less effective in scenarios with limited training samples or strict efficiency constraints.

As a result, it remains unclear whether traditional machine learning methods or lightweight CNN architectures provide a better trade-off between robustness, accuracy, and data efficiency for segmentation-based image focus detection. This research addresses this question through a systematic comparative study of both approaches.

2 Objective and Research Interest

This research investigates image focus detection based on a segmentation and classification framework. The main focus is to compare traditional machine learning methods and lightweight convolutional neural networks under conditions of limited training data.

Convolutional neural networks are widely recognized for their ability to automatically learn discriminative features, but they typically require a large number of labeled samples. Traditional machine learning approaches, in contrast, rely on handcrafted features and simpler models, which may be more suitable for small datasets and fast training. Since lightweight CNN architectures contain a limited number of parameters, it remains unclear whether they can consistently outperform classical methods in low-sample scenarios.

The objective of this study is to empirically evaluate and compare both approaches in terms of classification performance and efficiency when distinguishing focused and unfocused image regions. Through this comparative analysis, the research aims to identify under which conditions traditional machine learning methods are sufficient and when lightweight neural networks provide observable advantages for image focus detection.

3 Literature Review / Theoretical Background

Image focus detection aims to distinguish sharp regions from blurred regions within an image, which is important for tasks such as image enhancement, object recognition, and quality assessment. Early approaches mainly relied on handcrafted image features, such as edge strength, gradient magnitude, or local texture, combined with classical machine learning classifiers.[2] These methods are efficient and interpretable, but their performance depends heavily on the quality of feature design and can degrade under varying image conditions.

Common traditional classifiers include[1]:

Support Vector Machines (SVMs) separate data points by finding the optimal hyperplane that maximizes the margin between classes. They are effective in high-dimensional spaces and small datasets. SVMs work well when classes are linearly separable or can be transformed to be so with kernel functions.

Decision Trees classify samples by recursively splitting the data based on feature thresholds. Each internal node represents a feature decision, and each leaf node represents a class label. They are intuitive and easy to interpret but can overfit if the tree grows too deep.

Random Forests are ensembles of decision trees that combine predictions from multiple trees. They reduce overfitting and improve generalization through majority voting (for classification) or averaging (for regression). Random forests are robust to noisy data and can handle high-dimensional features.

Clustering Methods group similar data points based on a similarity metric, without using labeled data. They are often used for unsupervised segmentation or region grouping in images. Popular algorithms include

K-means and hierarchical clustering.

Convolutional neural networks (CNNs) have recently become popular in image analysis due to their ability to automatically learn hierarchical features from raw data. CNN-based methods can be more robust than traditional approaches, but they usually require a large number of labeled samples and substantial computational resources.[3] Lightweight CNN architectures reduce the number of parameters and computational cost, but it is not clear whether they maintain an advantage over traditional methods when training data is limited.

This study focuses on comparing traditional machine learning methods and lightweight CNNs for segmentation-based focus detection, particularly under low-sample conditions. The goal is to understand the trade-offs between robustness, accuracy, and efficiency, and to identify when each approach is most suitable.

4 Research Concept / Methodology

1. Research Focus

- Image focus detection as a classification problem:
 - 2-class: focused vs unfocused
 - 3-class: focused, edge, unfocused
- Compare traditional machine learning methods vs lightweight neural networks
- Focus on low-sample scenarios

2. Data Processing & Segmentation

- Divide images into patches / regions for classification
- Preprocessing: standardization, grayscale conversion
- Data augmentation for neural networks: rotation, flipping, noise addition

3. Traditional Machine Learning Methods

- **Algorithms:**
 - Support Vector Machine (SVM)[1]
 - Decision Tree[1]
 - Random Forest[1]
 - Optional: Clustering (for feature exploration / unsupervised analysis)
- **Features:**
 - Edge strength, gradients, Laplacian
 - Texture descriptors (GLCM, LBP[2])
 - Frequency-domain features (FFT, DCT)
- Advantages: fast training, interpretable
- Limitations: dependent on handcrafted features, limited robustness

4. Neural Network Methods

- **Algorithms:**
 - Lightweight Convolutional Neural Network (CNN)[3]
 - Optional: Transformer (attention mechanism for edge/blur focus)
- Characteristics:
 - Automatic feature extraction
 - Potentially higher robustness
 - Sensitive to small sample sizes, mitigated by lightweight architecture & augmentation

5. Experimental Design & Comparison

- Train both traditional and neural network models on the same dataset.
- Evaluate using multiple metrics:
 - Classification accuracy (2-class / 3-class)
 - Precision, Recall, and F1-score
 - Confusion matrix
 - Robustness across different image types and blur levels
 - Training efficiency and data requirements
- Variables to investigate:
 - Sample size
 - Classification type (binary vs multi-class)
 - Model type (traditional vs neural network)

6. Expected Outcomes

- Identify conditions where traditional ML is sufficient
- Determine scenarios where lightweight neural networks provide clear advantages
- Provide practical guidance for method selection in focus detection tasks

5 Preliminary Outline

1. Introduction and Research Motivation

- Background: importance and applications of image focus detection (image enhancement, object recognition, quality assessment, etc.)
- Research problem: performance comparison of traditional machine learning and lightweight neural networks under low-sample conditions
- Research objectives and contributions
- Thesis outline: briefly describe the structure of the thesis, including theoretical background, methodology, experiments, results, and conclusions

2. Theoretical Background

- **Traditional Machine Learning Methods**
 - Algorithms: SVM, Decision Tree, Random Forest, Clustering (optional)
 - Feature extraction:
 - * Edge strength, gradient, Laplacian
 - * Texture descriptors (GLCM, LBP)
 - * Frequency-domain features (FFT, DCT)
 - Implementation and training procedure
 - Evaluation strategy for 2-class and 3-class classification
- **Lightweight Neural Network Methods**
 - Algorithms: Lightweight CNN (mandatory), Transformer (optional)
 - Architecture and training details
 - Data augmentation and strategies for low-sample training
 - Evaluation strategy for 2-class and 3-class classification

3. Methodology: Image Segmentation and Preprocessing

- Divide images into patches or regions for classification
- Preprocessing: standardization, grayscale conversion
- Mapping classified patches back to the original image (visualization / optional)
- Data augmentation techniques for neural network training: rotation, flipping, noise addition

4. Experiments and Performance Evaluation

- Compare traditional ML vs lightweight neural networks
- Variables:
 - Sample size (low vs medium)
 - Classification type (2-class vs 3-class)
- Performance metrics:
 - Accuracy, Precision, Recall, F1-score
 - Confusion matrix
 - Robustness across different image types and blur levels
 - Training efficiency and computational cost
- Summary of comparative results

5. Discussion, Practical Significance and Future Work

- Analysis of strengths and limitations of each method
- Practical applications of the research findings
- Potential improvements and directions for future work

6 Thesis Timeline

1. **Week 1:** Literature review & theoretical background, finalize methods (traditional ML & lightweight CNN)
2. **Week 2:** Data preparation, image segmentation & preprocessing, data augmentation experiments
3. **Week 3:** Implement traditional ML methods, training and preliminary testing
4. **Week 4:** Implement lightweight CNN, low-sample training, preliminary performance comparison
5. **Week 5:** Full experiments (2-class / 3-class), record metrics (Accuracy, Precision, Recall, F1-score, Confusion Matrix)
6. **Week 6:** Analyze experiment results, create figures and tables, write methodology and experiments sections
7. **Week 7:** Complete thesis first draft (Introduction, Theoretical Background, Methodology, Experiments & Results)
8. **Week 8:** Thesis revision & polishing, check formatting and references, prepare for submission

Note: If a given week's tasks require more time, the week may be extended by one week, with subsequent tasks adjusted accordingly.

References

- [1] Kevin P. Murphy. *Probabilistic Machine Learning: An Introduction*. The MIT Press, Cambridge, MA, 2022.
- [2] Xin Yi and Mark G. Eramian. Lbp-based segmentation of defocus blur. *IEEE Transactions on Image Processing*, 25(4):1626–1638, 2016.
- [3] Kai Zeng, Yaonan Wang, Jianxu Mao, Junyang Liu, Weixing Peng, and Nankai Chen. A local metric for defocus blur detection based on cnn feature learning. *IEEE Transactions on Image Processing*, 28(5):2107–2115, 2019.